JY-ME02-485 Product Manual



Product Specifications:SPECIFICATION

Model: JY-ME02-485

Description: Angle encoder sensor

Production execution standard reference

Enterprise quality system standard: ISO9001:2016 standard

Sensor production standard: GB/T191SJ 20873-2016

Product testing standard: GB/T191SJ 20873-2016

1. Product Overview

- Encoders can measure angle, angular velocity, number of revolutions and temperature.
- The encoder has an angle measurement range from 0° to 360° and a revolution range from -32768 to 32767.
- The encoder has a linearity of 0.1% and an angular accuracy of ±0.01 degrees.
- The encoder can set the angle and angular velocity, and save the current angle and number of revolutions when power is off.
- The encoder has low resistance, small size, light weight, low power consumption and good mechanical installation compatibility.

2. Performance parameters

1. Voltage: 5V~36V

2. Current: <25mA

3. Volume: 55.5mm x 38mm

4. Data interface: 485 interface

5. Angle range: 0° to 360°

6. Angle stability: ±0.01°

7. Resolution: 15BIT

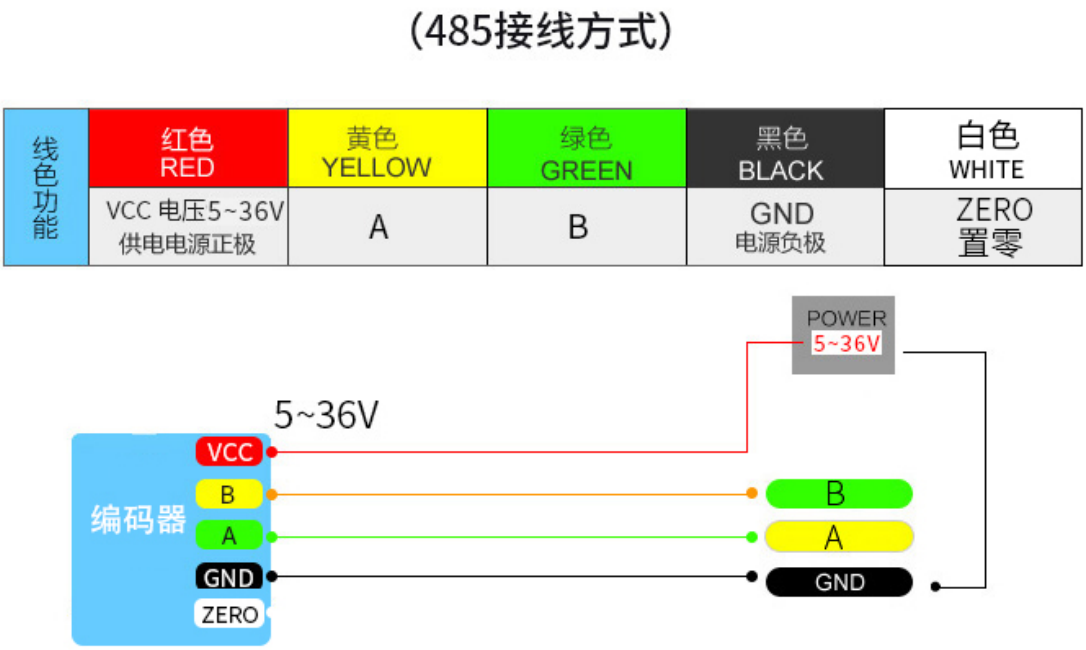
3. Pin function

Line Color	RED	YELLOW	GREEN	BLACK	WHITE
name	VCC 5-36V	485 A	485 B	GND	ZERO

Function	Positive power supply	485 A	485 B	Negative pole of power supply	Zero

Note: Connect the white wire and GND for two seconds to set the angle to zero.

4. Hardware connection method



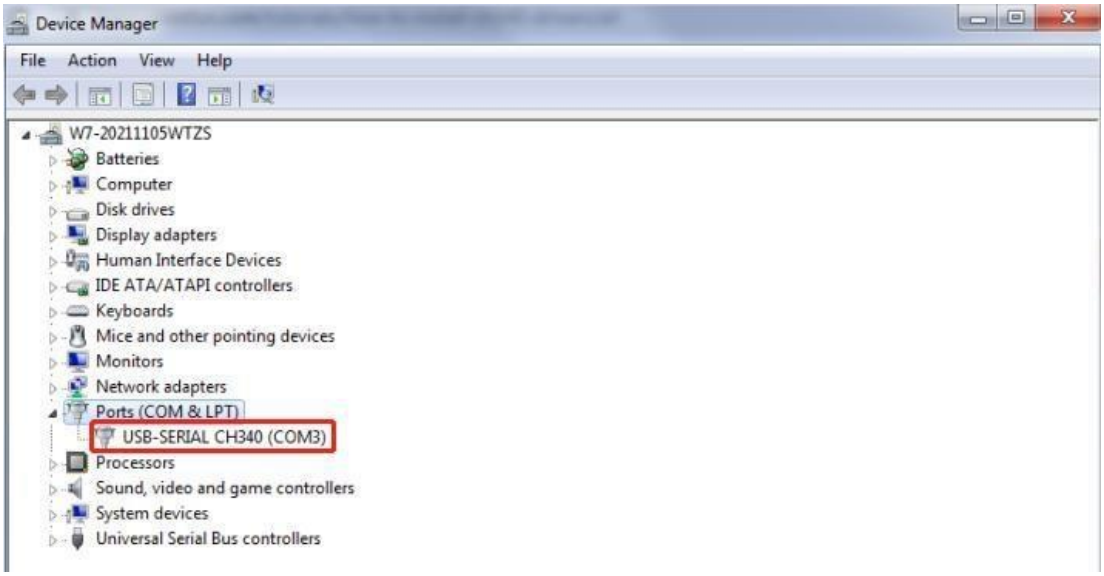
5. How to use the software

5.1 CH340 driver installation

Note: If the host computer cannot run, please download and install .net framework4.0:

Connect the USB to CAN module to the computer, turn on the host computer, install the CH340 driver corresponding to the serial port module https://drive.google.com/file/d/1JidopB42R9EsCzMAYC3Ya9eJ8JbHapRF/view?usp=drive_link

, and then query the corresponding port number in the device manager, as shown in the figure:



After the device is connected, open the JY-ME02 host computer software (the software can be downloaded from https://drive.google.com/file/d/10xysnkuyUwi3AK_t3965SLr5Yt6YKE-u/view?usp=drive_link

, here we take JY-ME02-485 as an example);

1. Search for the corresponding model.

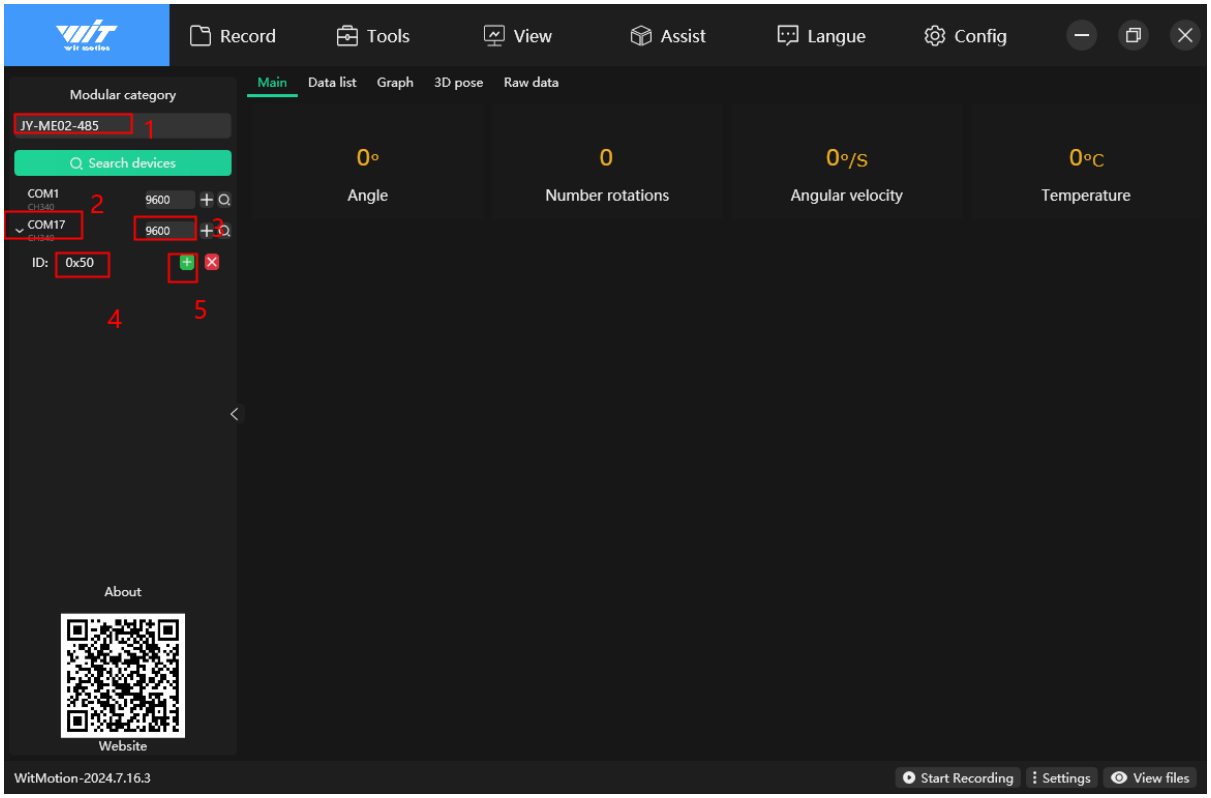
2. Select the corresponding port.

3. Select the baud rate of the module (the default baud rate of 485 baud rate of JY-ME02 is 9600).

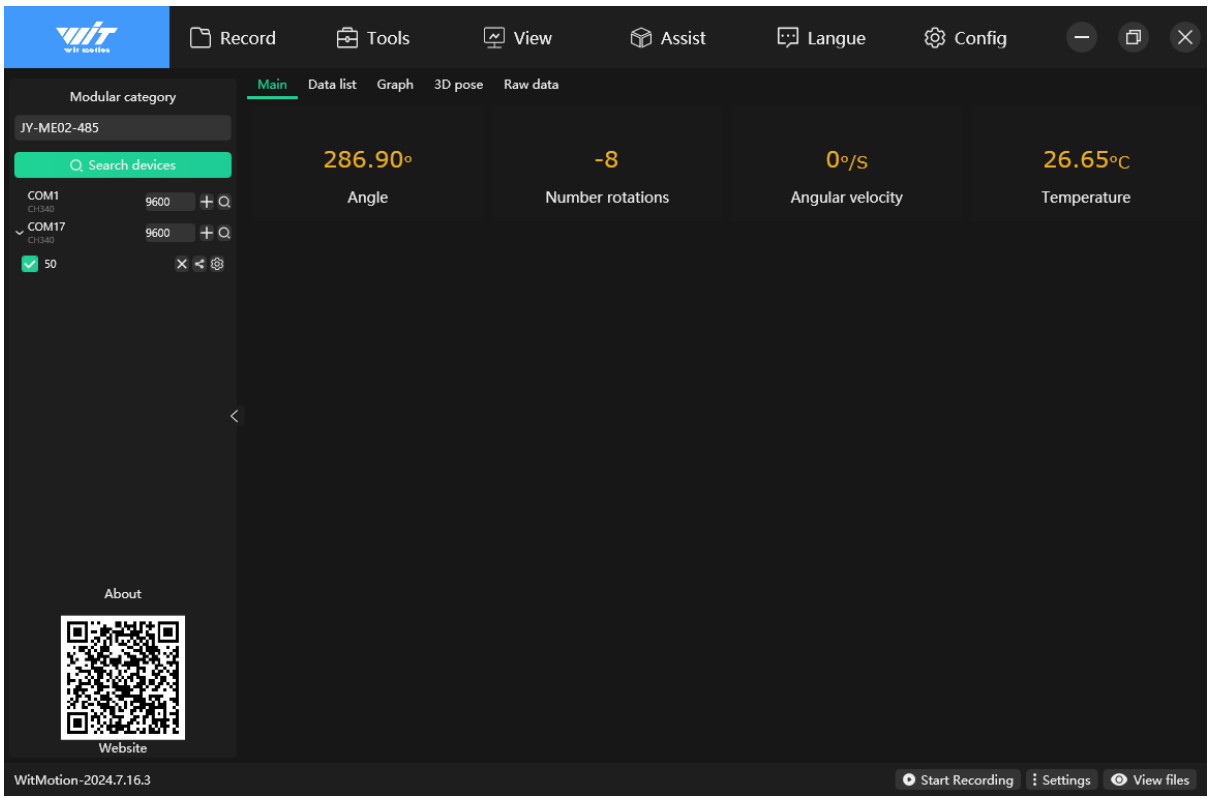
4. Enter the corresponding ID (the default is 0x50).

5. Click Add Device

As shown below:

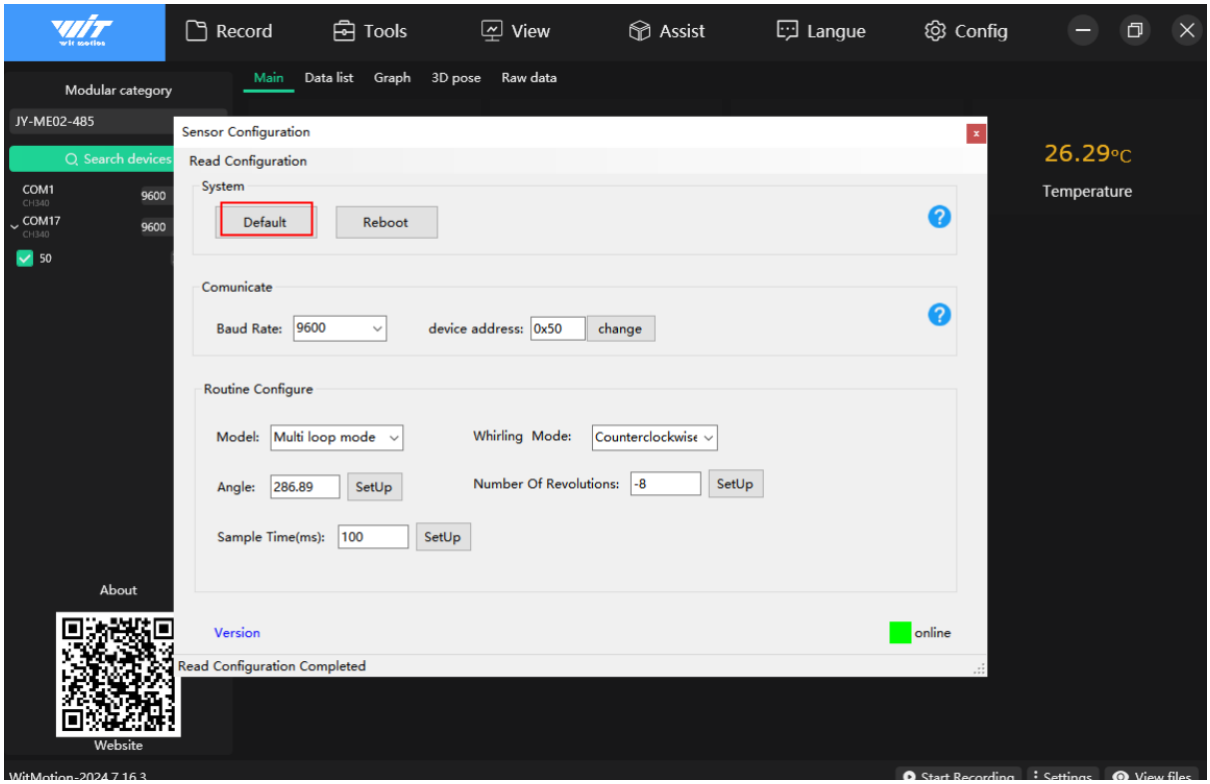


After the settings are completed, the data will appear on the software, as shown below:



5.2 Restore factory settings

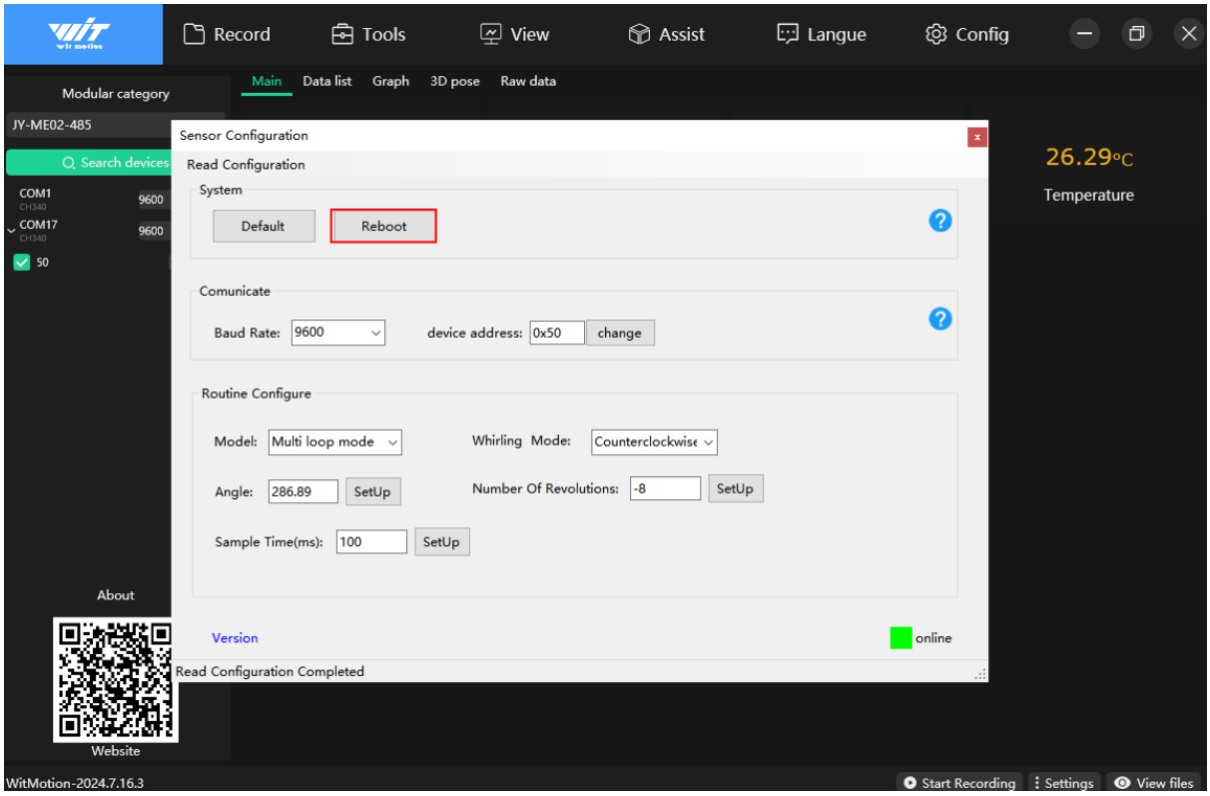
Click "Configuration" and then click "Restore Settings" in the sensor configuration interface to restore the factory settings, as shown in the following figure:



Zero line to restore factory settings: Before powering on the encoder, first ground the zero line (white), then power on, wait for two seconds, and finally power on again to restore the factory settings.

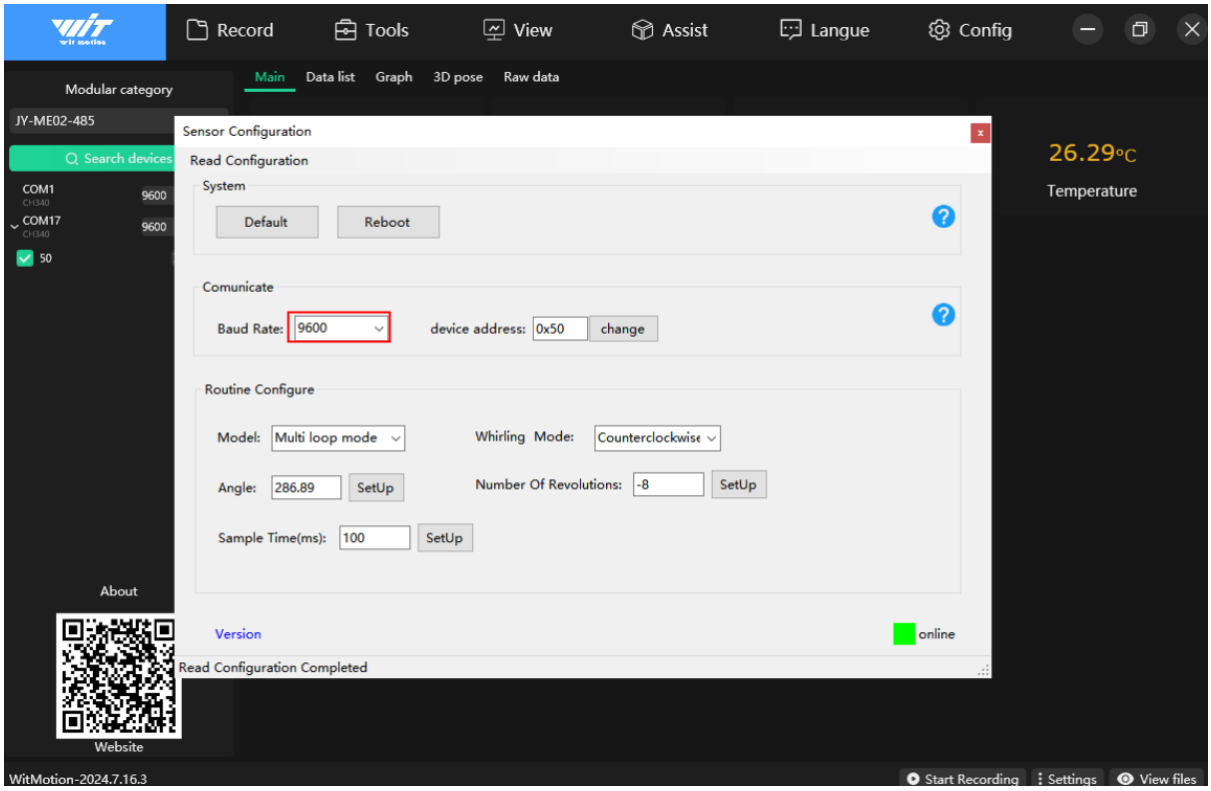
5.3 Restart

Click "Configure", and then click "Restart" in the sensor configuration interface to restart the sensor, as shown in the following figure:



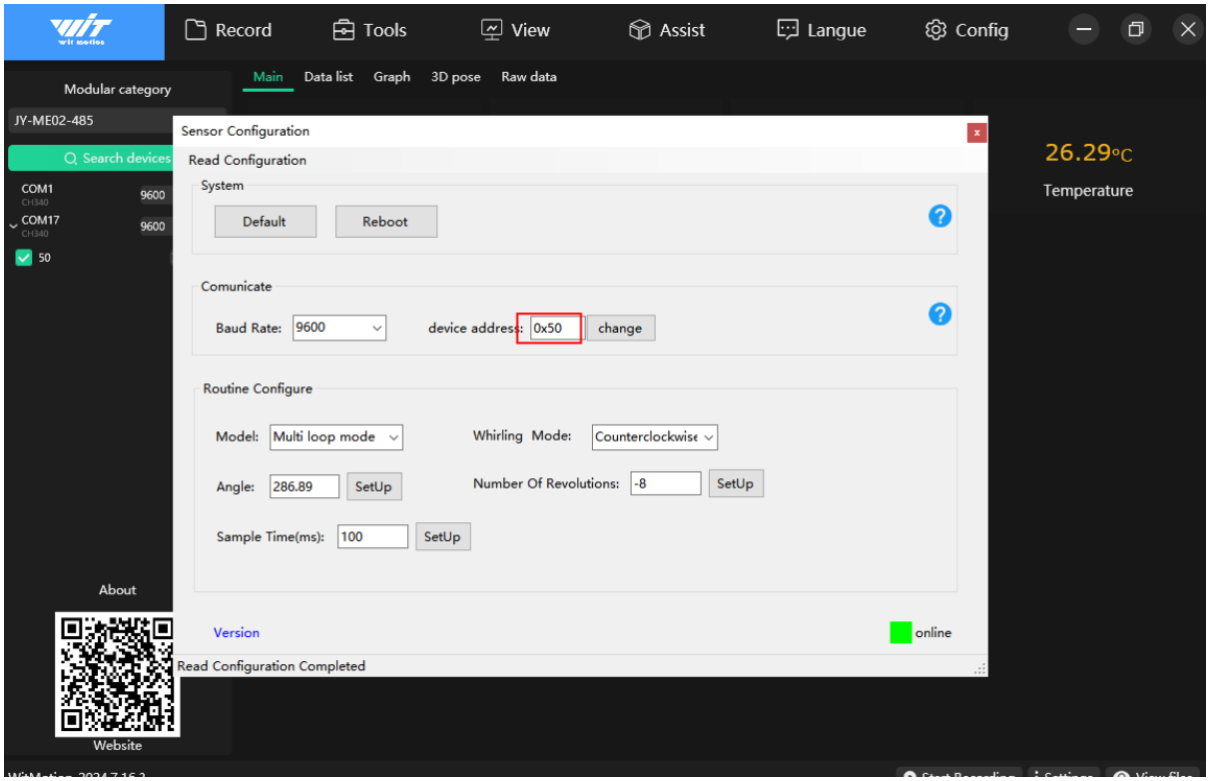
5.4 Change the serial port baud rate

Open "Configuration", click the drop-down menu of "Serial Port Baud Rate" in the sensor configuration interface, select the serial port baud rate to be modified, and you can change the current serial port baud rate (the default serial port baud rate is 9600). The serial port baud rate can be 4800, 9600, 19200, 38400, 57600, 115200 and 230400, as shown in the following figure:



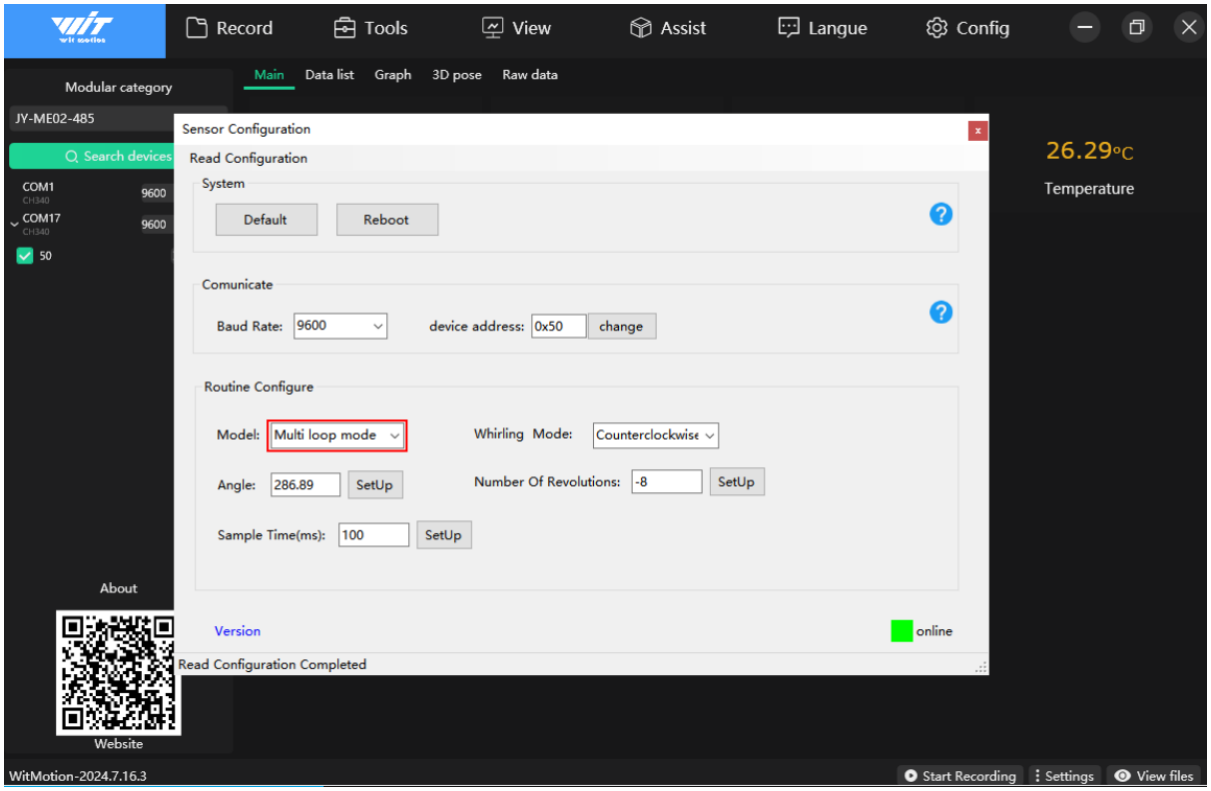
5.5 Change Modbus address

Open "Configuration", click the "Modbus Address" input box in the sensor configuration interface, enter the Modbus address and then click Set to change the Modbus address (the default Modbus address is 0x50). The Modbus address ranges from 0x00 to 0x7F, as shown in the figure:



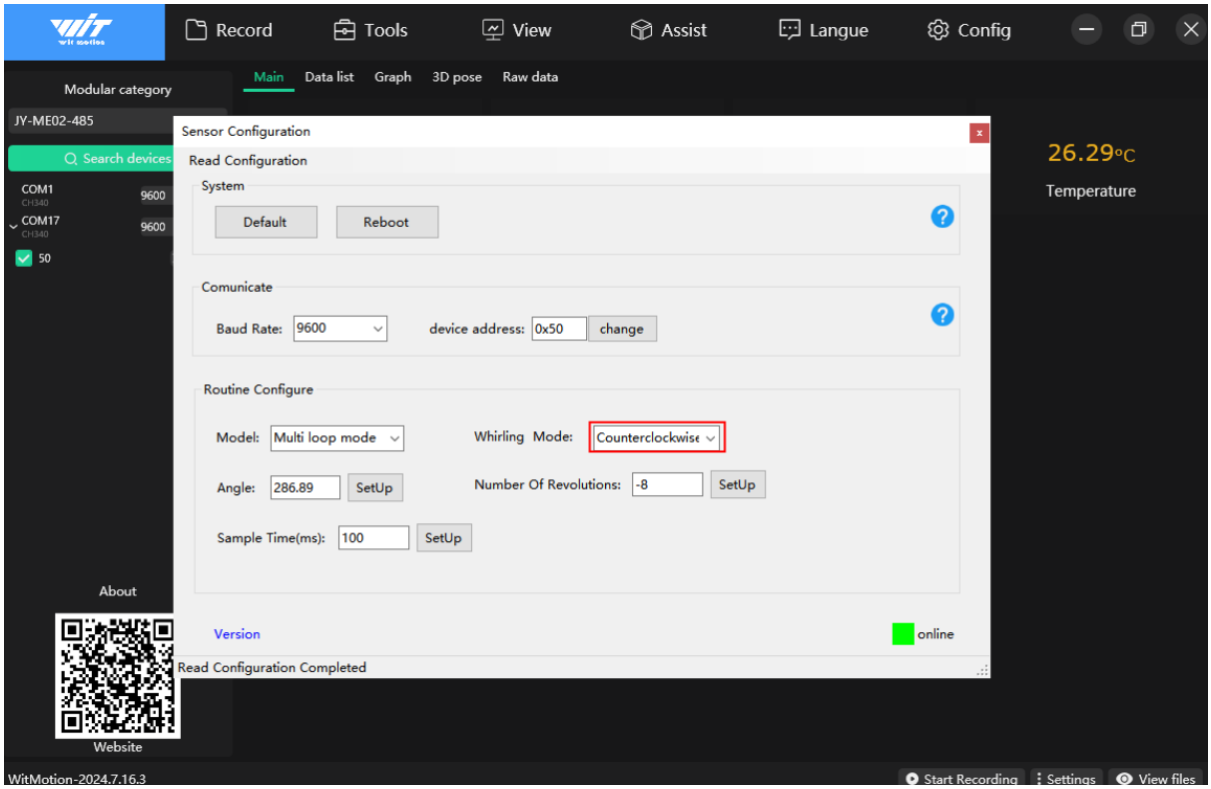
5.6 Setting Mode

Open "Configuration" and click the "Mode" drop-down box in the sensor configuration interface to set the current encoder mode to single-turn mode or multi-turn mode. The revolution output in single-turn mode is 0. In multi-turn mode, the current revolution will be counted, and the counting revolution range is from -32768 to 32767, as shown in the figure:



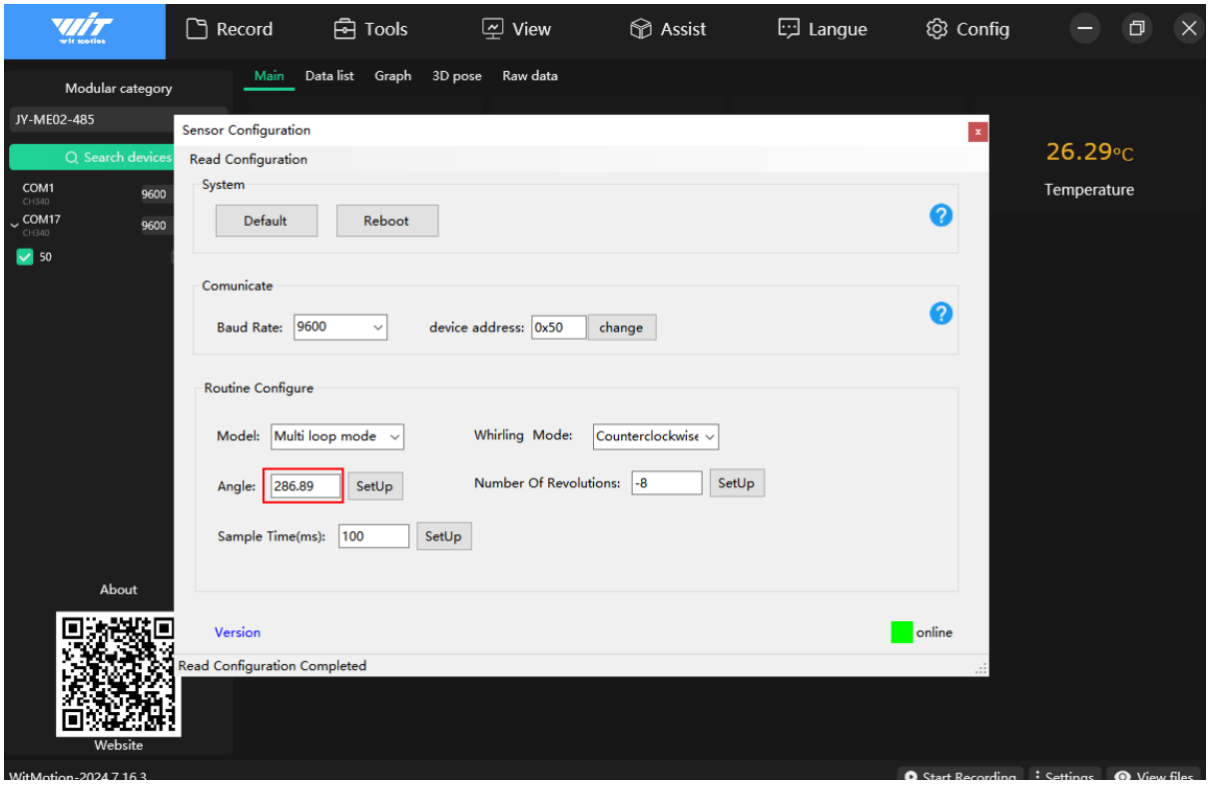
5.7 Setting the rotation direction

Open "Configuration" and click the "Mode" drop-down box in the sensor configuration interface to set the rotation mode. When the clockwise rotation mode is set, the sensor will count positively when it rotates clockwise. When the counterclockwise rotation mode is set, the sensor will count negatively when it rotates clockwise. As shown in the figure:



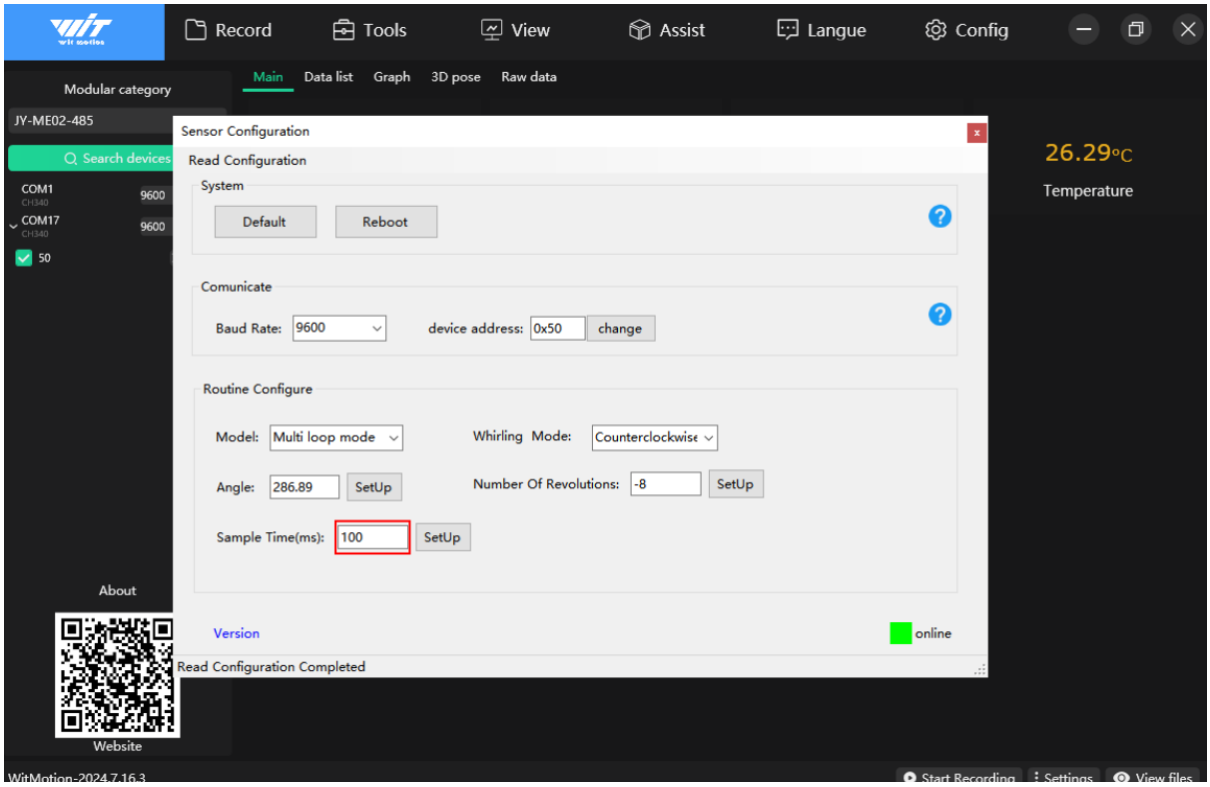
5.8 Setting the Angle

Open "Configuration", click the "Angle" input box in the sensor configuration interface, enter the angle to be set and then click Set. The settable angle ranges from 0° to 360°, as shown in the figure:



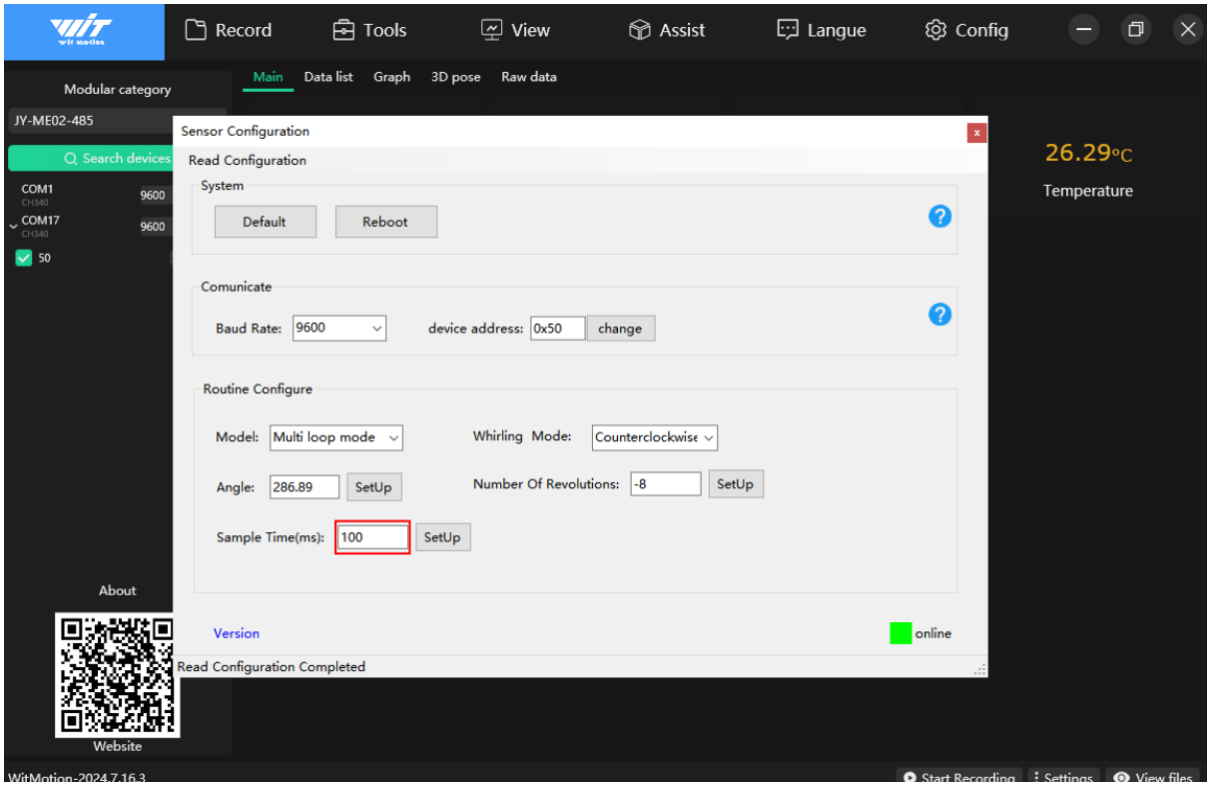
5.9 Setting the number of revolutions

Open "Configuration", click the "Revolutions" input box in the sensor configuration interface, enter the revolutions to be set and click Set. The revolutions that can be set range from -32768 to 32767, as shown in the figure:



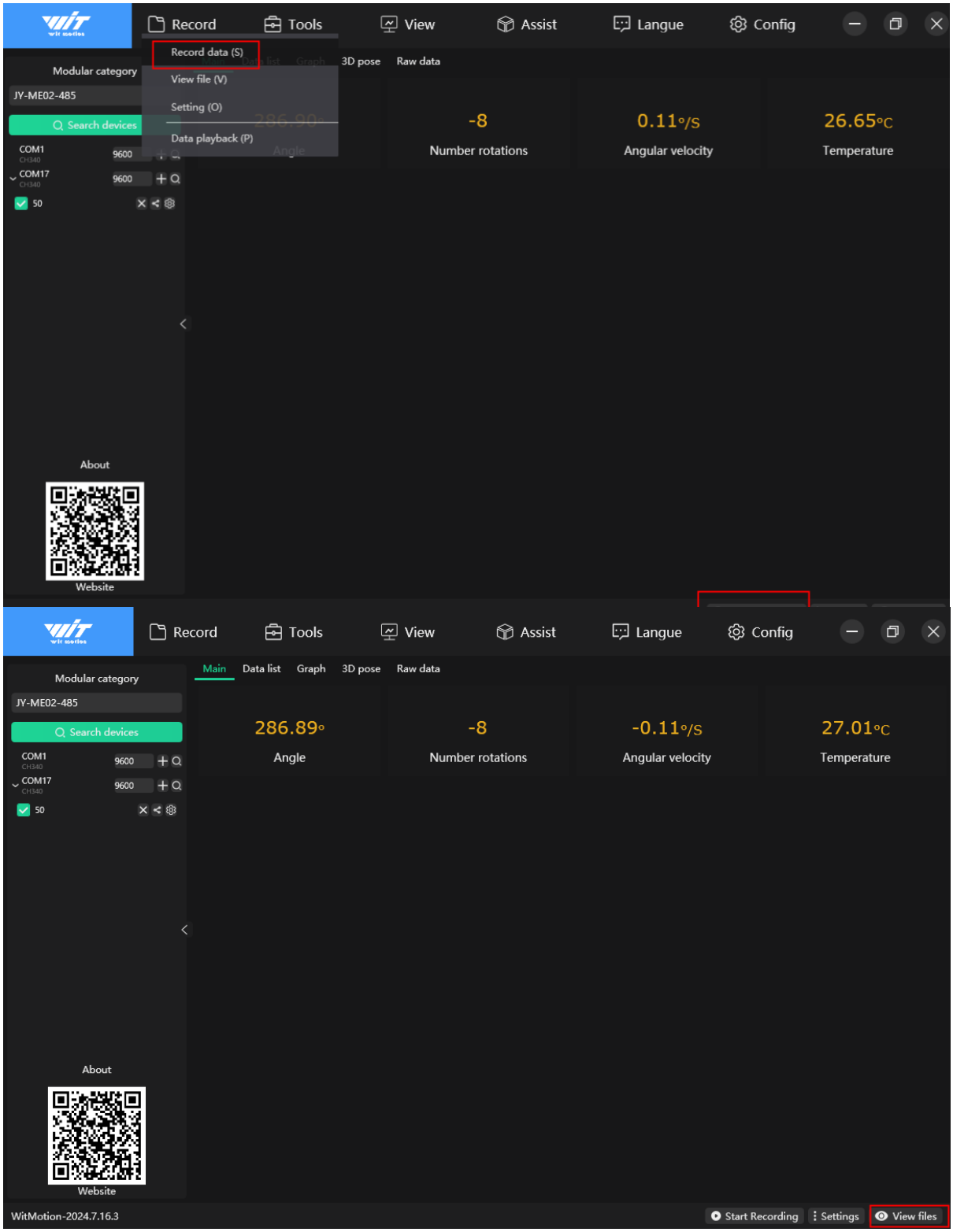
5.10 Setting the angular velocity sampling time

Open "Configuration", click the "Sampling Time" input box in the sensor configuration interface, enter the angular velocity to be set, and then click Set. The calculation method of angular velocity is the angle of change divided by the angle of change time. The angle of change is equal to the value of the next moment minus the value of the previous moment. This is automatically calculated internally, and the time that needs to change now is the sampling time. The encoder sampling time defaults to 100ms, and the default value of the encoder sampling time register is 1000 (the encoder sampling time register is 1, which is 100us), that is, the angular velocity is calculated once every 100ms. The value of the angular velocity sampling time register can be set from 1 to 10000, and the angular velocity sampling time that the host computer can set is 1ms to 1000ms, as shown in the figure:



5.11 Recording Data

Open "Record" and click "Start Recording" to record the output data of the encoder.



The saved file begins with the values corresponding to the data, the first column: Time, second column: serial port number, third column: angle, fourth column:

number of revolutions, fifth column: angular velocity, sixth column: temperature.

Time	Device name	Angle(°)	Number rotati	Angular veloc	Temperature(° C
17:28:30.767	COM1750	286.89	-8	0	25.93
17:28:30.923	COM1750	286.89	-8	0	26.65
17:28:31.075	COM1750	286.89	-8	0	25.93
17:28:31.229	COM1750	286.89	-8	0	27.01
17:28:31.381	COM1750	286.89	-8	0	25.93
17:28:31.533	COM1750	286.89	-8	0	27.01
17:28:31.684	COM1750	286.89	-8	0	25.93
17:28:31.836	COM1750	286.89	-8	0	26.65
17:28:31.987	COM1750	286.89	-8	0	25.93
17:28:32.140	COM1750	286.89	-8	0	26.65
17:28:32.292	COM1750	286.89	-8	0	25.93
17:28:32.444	COM1750	286.86	-8	-0.11	27.01
17:28:32.596	COM1750	286.88	-8	-0.11	25.93
17:28:32.747	COM1750	286.88	-8	0	27.01
17:28:32.899	COM1750	286.86	-8	0	25.93
17:28:33.050	COM1750	286.88	-8	0.22	27.01
17:28:33.203	COM1750	286.89	-8	0	25.93
17:28:33.355	COM1750	269.82	-8	-124.805	25.93
17:28:33.507	COM1750	294.84	-8	-32.3	27.01
17:28:33.660	COM1750	174.5	-8	-835.181	25.93
17:28:33.812	COM1750	140	-8	-631.604	27.01
17:28:33.965	COM1750	160.53	-8	193.799	25.93
17:28:34.116	COM1750	155.76	-8	-47.351	26.65
17:28:34.268	COM1750	48.32	-8	-781.238	26.65
17:28:34.419	COM1750	20.05	-8	-98.657	25.93
17:28:34.572	COM1750	29.5	-8	35.266	26.65
17:28:34.725	COM1750	58.68	-8	207.532	25.93
17:28:34.877	COM1750	346.97	-9	-325.415	26.65
17:28:35.030	COM1750	272.46	-9	-491.748	25.93
17:28:35.181	COM1750	266.74	-9	-72.729	26.65
17:28:35.334	COM1750	259.79	-9	-67.236	26.65
17:28:35.486	COM1750	238.39	-9	-89.648	25.93
17:28:35.637	COM1750	168.85	-9	-524.707	27.01
17:28:35.788	COM1750	158.98	-9	-260.266	27.01
17:28:35.939	COM1750	159.15	-9	1.208	25.93
17:28:36.092	COM1750	160.23	-9	55.591	27.01
17:28:36.243	COM1750	44.35	-9	-997.119	25.93
17:28:36.395	COM1750	358.85	-10	-319.043	27.01
17:28:36.547	COM1750	298.5	-10	-334.753	25.93
17:28:36.699	COM1750	271.49	-10	-436.926	26.65
17:28:36.851	COM1750	216.21	-10	-166.772	26.65
17:28:37.003	COM1750	179.89	-10	-120.959	26.65

6. Communication Protocol

Level: 485 level (default baud rate: 9600)

6.1 Protocol Description

The format of the read command sent by the host computer is as follows:

ID	Command number	Register address high	Register address low	Read length high	Read length low	CRC check high bit	CRC check low bit
ID	0x03	RegH	RegL	LenH	LenL	CRCH	CRCL

The lower computer responds to the read instruction format as shown in the following table:

ID	Command number	Data length	The first register data high bit	The first register data low	The second register data high	The second register data low	...	...	CRC check high bit	CRC check low bit
ID	0x03	DataLen	Reg1H	Reg1L	Reg2H	Reg2L	...	...	CRCH	CRCL

The format of the write command sent by the host computer is as follows:

ID	Command number	Register address high	Register address low	Register value high	Register value low	CRC check high bit	CRC check low bit
ID	0x06	RegH	RegL	V L	V L	CRCH	CRCL

6.2 Register Address Table

address	symbol	meaning
0x00	SAVE	save

0x04	BAUD	Baud rate
0x10	ENCODER_MODE	Encoder Mode
0x11	ANG_VAL	Angle value
0x12	REVOLUTI	Revolutions
0x13	ANGSPE_VAL	Angular velocity
0x14	TEMP	temperature
0x15	SPIN_DIR	turn around
0x17	ANGSPE_SAMPLING_TIME	Angular velocity sampling time
0x1A	DEV_ADDR	Device Address
0x2E	VERSIONL	Version number low
0x2F	VERSION H	Version number high

7. Communication instructions

7.1 Write Instructions

7.1.1 Setting Unlock

0x50	0x06	0x00	0x69	0xB5	0x88	CRCH	CRCL

Each time before using the command to set the encoder, you must send the unlock command, otherwise the command to set the sensor will fail.

7.1.2 Save settings, restore default (factory) configuration and save, restart

0x50	0x06	0x00	0x00	0x00	SAVE	CRCH	CRCL

SAVE: Save the value of the register

0x00: Save the current configuration

0x01: Restore the default (factory) configuration and save it

0xFF: Restart

7.1.3 Setting the baud rate

0x50	0x06	0x00	0x04	0x00	BAUD	CRCH	CRCL

BAUD: baud rate register value

0x01:4800

0x02: 9600 (default)

0x03:19200

0x04:38400

0x05:57600

0x06:115200

0x07:230400

7.1.4 Set Modbus address

0x50	0x06	0x00	0x1A	DEV_ADDR_H	DEV_ADDR_L	CRCH	CRCL

DEV_ADDR_L: Modbus address register low

DEV_ADDR_H: Modbus address register high

DEV_ADDR = DEV_ADDR_H <<8 | DEV_ADDR_L

Modbus address is 0x50, the minimum Modbus address is 0x00, and the maximum Modbus address is 0x7F.

7.1.5 Setting Mode

0x50	0x06	0x00	0x15	0x00	MODE	CRCH	CRCL

MODE: value of the mode register

0x00: Set the encoder mode to single turn mode

0x01: Set the encoder mode to multi-turn mode

7.1.6 Setting the rotation direction

0x50	0x06	0x00	0x15	0x00	SPIN_DIR	CRCH	CRCL

SPIN_DIR: value of the rotation mode register

0x00: Set the encoder rotation mode to clockwise

0x01: Set the encoder rotation mode to counterclockwise

7.1.7 Setting the Angle

The encoder angle formula is: Angle (°) = ANG_VAL * 360 / 32768.

example:

Set the encoder angle to 30°. ANG_VAL = 30 * 32768 / 360 = 2731. The decimal number 2731 corresponds to the hexadecimal number 0x0AAB.

0x50	0x06	0x00	0x11	0x0A	0xAB	CRCH	CRCL

7.1.8 Setting the angular velocity sampling time

0x50	0x06	0x00	0x17	SAMPL_TIME_H	SAMPL_TIME_L	CRCH	CRCL

SAMPL_TIME_L: Angular velocity sampling time register low

SAMPL_TIME_H: Angular velocity sampling time register high

SAMPL_TIME = SAMPL_TIME_H << 8 | SAMPL_TIME_L

Angular velocity sampling time = SAMPL_TIME * 100us

The default value of the angular velocity sampling time register is 1000, and the corresponding angular velocity sampling time is 100ms. The minimum value of the angular velocity sampling time register is 1, and the corresponding angular velocity sampling time is 100us. If the encoder rotates at a fast speed, it is easy to exceed the range when calculating the value of the angular velocity register internally (the angular velocity register has only 16 bits), resulting in problems with the calculated angular velocity value. The value of the angular velocity sampling time register can be set to a small value, so that the value of the angular velocity register will not exceed the range, thereby preventing calculation errors.

7.2 Read Instructions

7.2.1 Reading baud rate

The host computer sends instructions

0x50	0x03	0x00	0x04	0x00	0x01	CRCH	CRCL

The lower computer responds to the data

0x50	0x03	0x02	0x00	BAUD	CRCH	CRCL

BAUD: baud rate register value

0x01:4800

0x02: 9600 (default)

0x03:19200

0x04:38400

0x05:57600

0x06:115200

0x07:230400

7.2.2 Read Modbus address

The host computer sends instructions

0x50	0x03	0x00	0x1A	0x00	0x01	CRCH	CRCL

The lower computer responds to the data

0x50	0x03	0x02	DEV_ADDR_H	DEV_ADDR_L	CRCH	CRCL

DEV_ADDR_L: Modbus address register low

DEV_ADDR_H: Modbus address register high

DEV_ADDR = DEV_ADDR_H <<8 | DEV_ADDR_L

Modbus address is 0x50, the minimum Modbus address is 0x00, and the maximum Modbus address is 0x7F.

7.2.3 Read Mode

The host computer sends instructions

0x50	0x03	0x00	0x10	0x00	0x01	CRCH	CRCL

The lower computer responds to the data

0x50	0x03	0x02	0x00	MODE	CRCH	CRCL

MODE: value of the mode register

0x00: Set the encoder mode to single turn mode

0x01: Set the encoder mode to multi-turn mode

7.2.4 Reading the direction of rotation

The host computer sends instructions

0x50	0x03	0x00	0x15	0x00	0x01	CRCH	CRCL

The lower computer responds to the data

0x50	0x03	0x02	0x00	SPIN_DIR	CRCH	CRCL

SPIN_DIR: value of the rotation mode register

0x00: Set the encoder rotation mode to clockwise

0x01: Set the encoder rotation mode to counterclockwise

7.2.5 Reading Angle

The host computer sends instructions

0x50	0x03	0x00	0x11	0x00	0x01	CRCH	CRCL

The lower computer responds to the data

0x50	0x03	0x02	ANG_VAL_H	ANG_VAL_L	CRCH	CRCL

ANG_VAL_H: High bit of the angle register value

ANG_VAL_L: Low bit of the angle register value

ANG_VAL = ANG_VAL_H<<8 | ANG_VAL_L

The encoder angle formula is: Angle (°) = ANG_VAL * 360 / 32768.

7.2.6 Reading the number of revolutions

The host computer sends instructions

0x50	0x03	0x00	0x12	0x00	0x01	CRCH	CRCL

The lower computer responds to the data

0x50	0x03	0x02	REVOLUTI_H	REVOLUTI_L	CRCH	CRCL

REVOLUTI_H：High bit of the revolution register value

REVOLUTI_L：Low bit of the revolution register value

REVOLUTI = REVOLUTI_H <<8 | REVOLUTI_L

REVOLUTI is the number of revolutions

7.2.7 Reading angular velocity sampling time

The host computer sends instructions

0x50	0x03	0x00	0x17	0x00	0x01	CRCH	CRCL

The lower computer responds to the data

0x50	0x03	0x02	ANGSPE_SAMPLING_TIME_H	ANGSPE_SAMPLING_TIME_L	CRCH	CRCL

ANGSPE_SAMPLING_TIME_H: High bit of the value of the revolution register

ANGSPE_SAMPLING_TIME_L: Low bit of the value of the revolution register

ANGSPE_SAMPLING_TIME = ANGSPE_SAMPLING_TIME_H<<8 | ANGSPE_SAMPLING_TIME_L

ANGSPE_SAMPLING_TIME is the angular velocity sampling time. If the sampling time register value is 100, the sampling time is 10ms, that is, the sampling time = the value of the sampling time register * 100 us.

7.2.8 Reading angular velocity

The host computer sends instructions

0x50	0x03	0x00	0x13	0x00	0x01	CRCH	CRCL

The lower computer responds to the data

0x50	0x03	0x02	ANGSPE_VAL_H	ANGSPE_VAL_L	CRCH	CRCL

ANGSPE_VAL_H：High bit of the angular velocity register value

ANGSPE_VAL_L：Low bit of the angular velocity register value

ANGSPE_VAL = ANGSPE_VAL_H <<8 | ANGSPE_VAL_L

ANGSPE_VAL is the angular velocity, angular velocity (°/s) = angular velocity register value * 360 / 32768 / angular velocity sampling time (s). (Note: The angular velocity sampling time above is calculated in seconds, and the default is 0.1s)

7.2.9 Reading Temperature

The host computer sends instructions

0x50	0x03	0x00	0x14	0x00	0x01	CRCH	CRCL

The lower computer responds to the data

0x50	0x03	0x02	TEMP_H	TEMP_L	CRCH	CRCL

TEMP_H：Temperature register value high

TEMP_L：Low bit of the temperature register value

TEMP = TEMP_H << 8 | TEMP_L

TEMP is temperature, temperature (℃) = TEMP / 100.